0/1 Knapsack Problem

Select the best combination of items without exceeding the maximum knapsack capacity.

Knapsack Capacity

8

Available Items

Item Name	Weight	Value
Laptop	3	
Book	2	
Camera	4	
Phone	1	
pen	1	

Run Knapsack Algorithm

Optimization Results

Maximum Value
105

Total Weight
8 / 8

Selected Items
3

Optimal item combination found!

Item	Weight	Value
Laptop	3	
Camera	4	
Phone	1	

View Dynamic Programming Table

Each cell dp[i][w] represents the maximum value obtainable using the first i items with capacity w.

	Capacity 0	Capacity 1	Capacity 2	Capacity 3	Capacity 4	Capacity 5	Capacity 6	Capacity 7	Capaci
Items 0	0	0	0	0	0	0	0	0	
Items 1	0	0	0	40	40	40	40	40	
Items 2	0	0	25	40	40	65	65	65	
Items 3	0	0	25	40	50	65	75	90	
Items 4	0	15	25	40	55	65	80	90	
Items 5	0	15	25	40	55	65	80	90	